CS2201 Computer Organization & Architecture

L26: Introduction to DRAM System



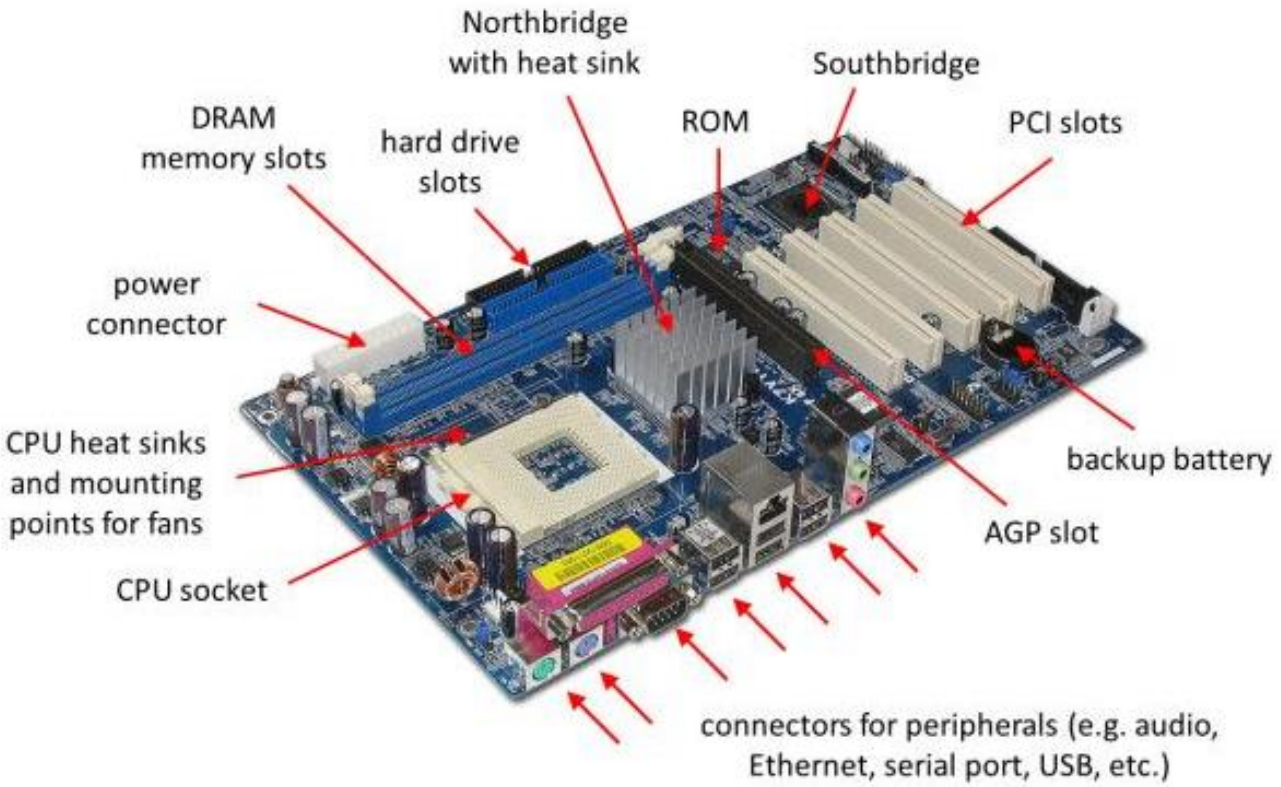
John Jose

Associate Professor

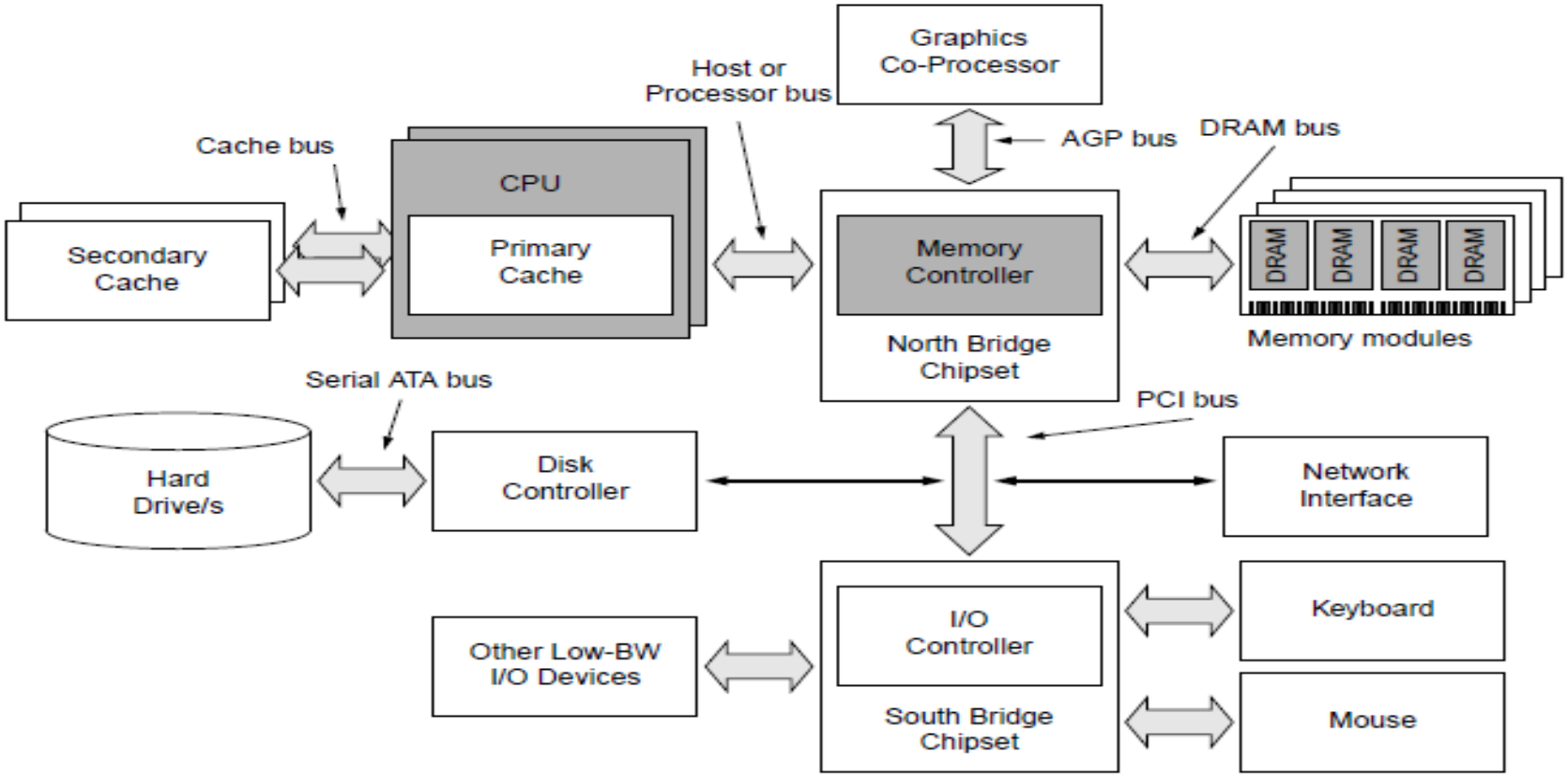
Department of Computer Science & Engineering

Indian Institute of Technology Guwahati

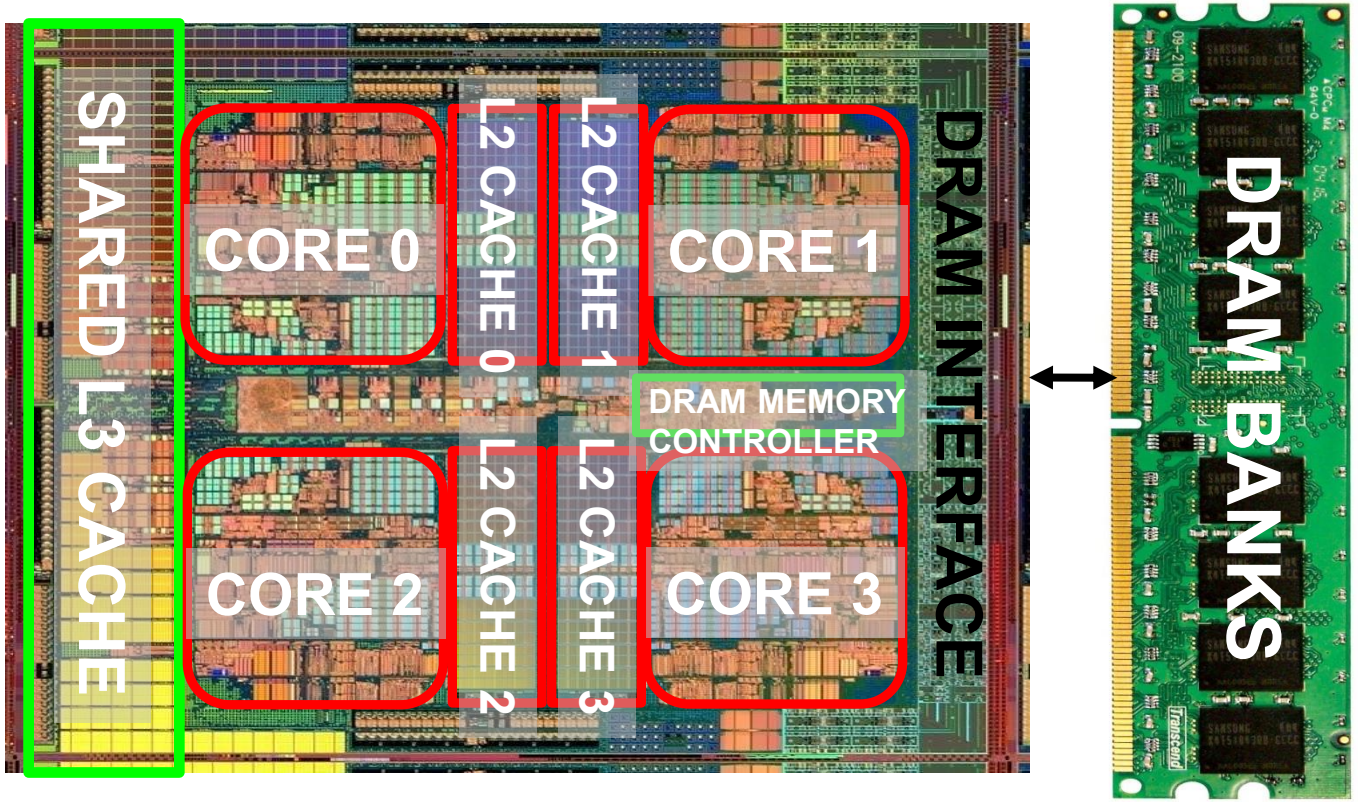
Components of a Modern Computer



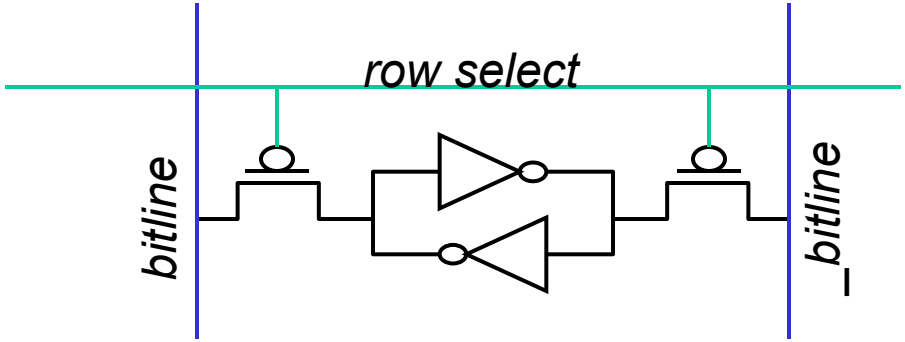
Components of a Modern Computer



Main Memory in the System

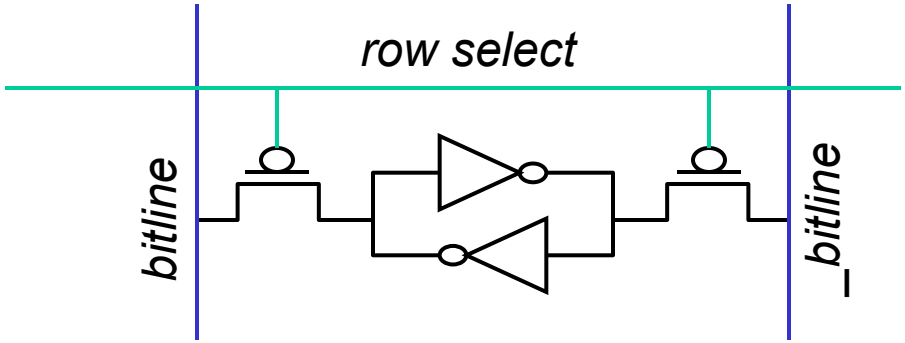


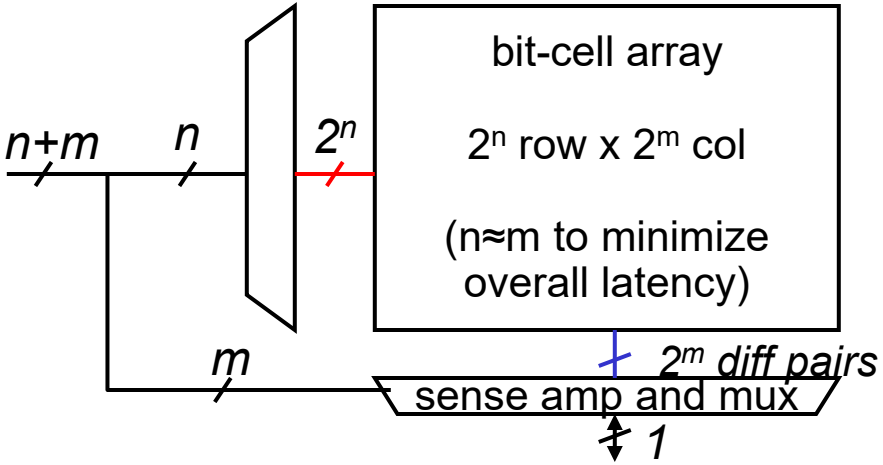
SRAM (Static Random Access Memory)



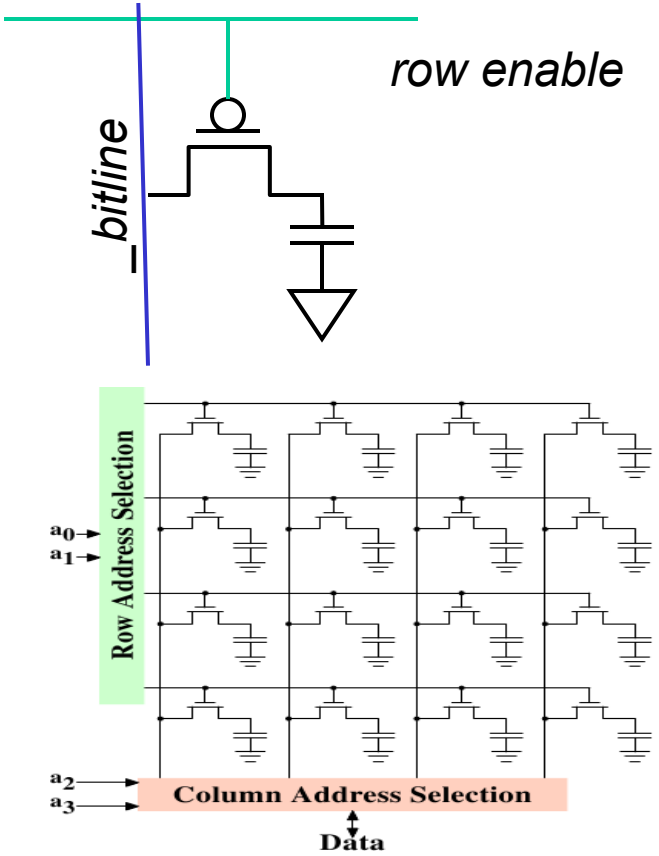
- **Read Sequence**
- address decode
- drive row select
- selected bit-cells drive bitlines (entire row is read together)
- Col. Select (data is ready)

SRAM (Static Random Access Memory)





DRAM (Dynamic Random Access Memory)



- Bits are stored as charges on capacitor
- B cell loses charge when read
- B cell loses charge over time
- A“flip-flopping” sense amplifier amplifies and regenerates the bitline, data bit is mux’ed out

DRAM vs SRAM

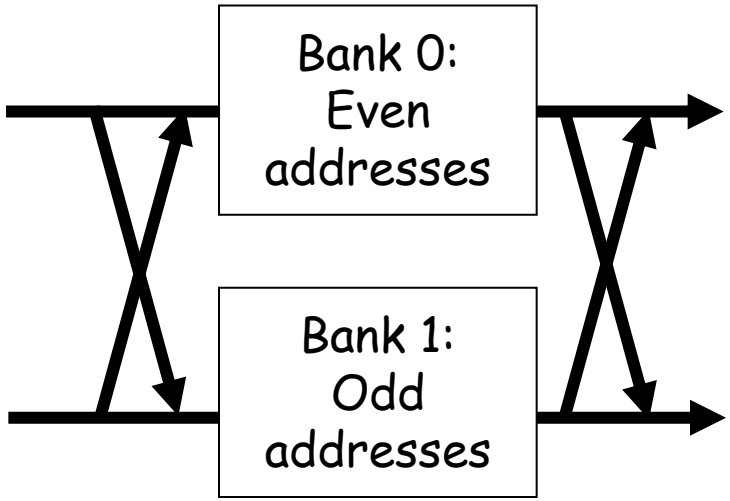
- **DRAM**
  - Slower access (capacitor)
  - Higher density (1T, 1C cell), Lower cost
  - Requires refresh (power, performance, circuitry)
  - Manufacturing requires putting capacitor and logic together
- **SRAM**
  - Faster access (no capacitor)
  - Lower density (6T cell), Higher cost
  - No need for refresh
  - Manufacturing compatible with logic process (no capacitor)

Principle of Interleaving

- **Banking (Interleaving)**
  - Problem: a single monolithic memory array takes long to access and does not enable multiple accesses in parallel
  - Goal: Reduce the latency of memory array access and enable multiple accesses in parallel
  - Idea: Divide the array into multiple banks that can be accessed independently (in the same cycle or in consecutive cycles)
    - Each bank is smaller than the entire memory storage
    - Accesses to different banks can be overlapped
  - A Key Issue: How do you map data to different banks?

Handling Multiple Accesses per Cycle

- **Banking (Interleaving)**
  - Address space partitioned into separate banks
  - No increase in data store area
  - Bits in address determines which bank an address maps to
  - Cannot satisfy multiple accesses to the same bank
  - Crossbar interconnect in input/output
  - Bank conflicts - Two accesses are to the same bank difficult to handle

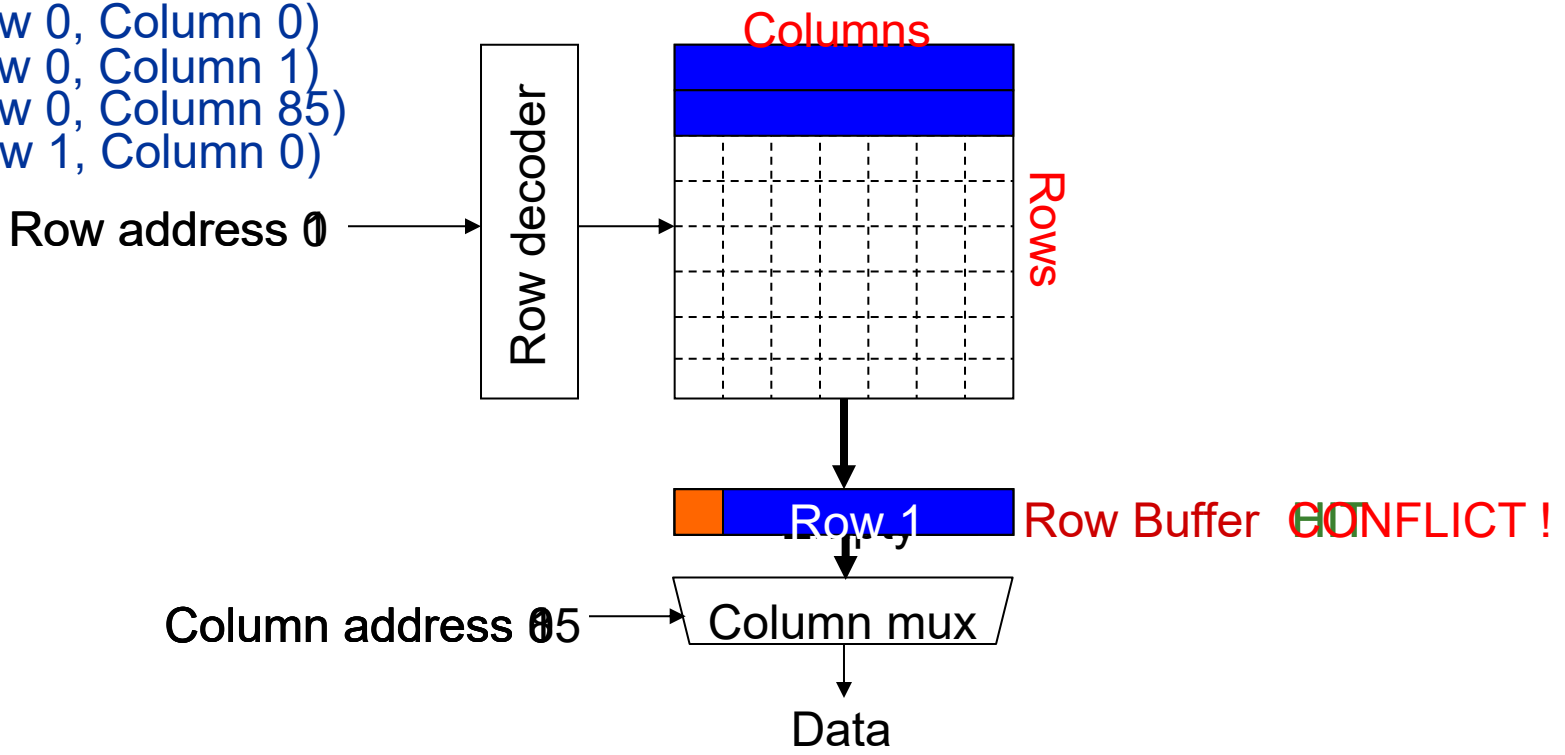


Page Mode DRAM

- A DRAM bank is a 2D array of cells: rows x columns
- Sense amplifiers are kept in row buffer
- Each address is a <row,column> pair
- **Access to a closed row**
  - Activate command opens row (placed into row buffer)
  - Read/write command reads/writes column in the row buffer
  - Precharge command closes the row and prepares the bank for next access
- **Access to an open row**
  - No need for activate command

DRAM Bank Operation

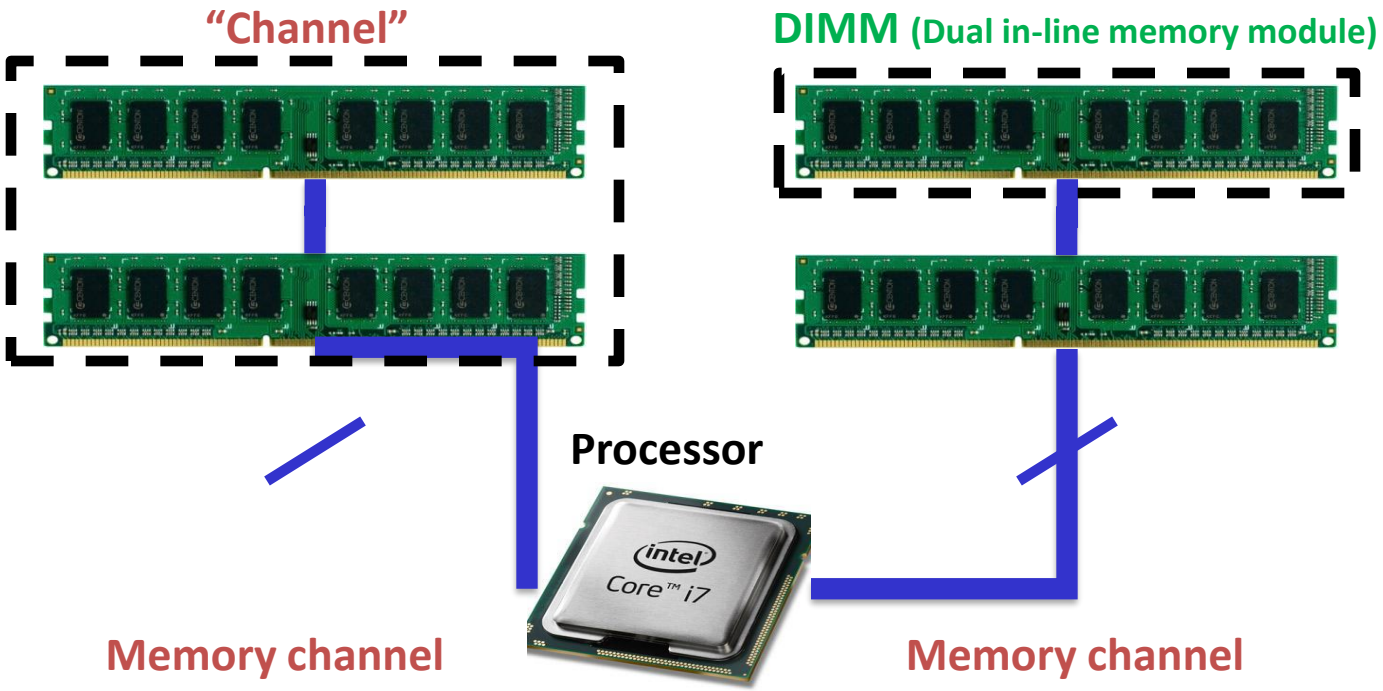
Access Address:
(Row 0, Column 0)
(Row 0, Column 1)
(Row 0, Column 85)
(Row 1, Column 0)



DRAM Subsystem Organization

- Channel
- DIMM
- Rank
- Chip
- Bank
- Row
- Column
- B-Cell

The DRAM subsystem

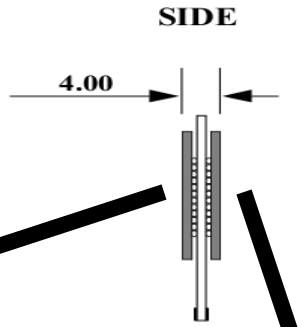


Breaking down a DIMM

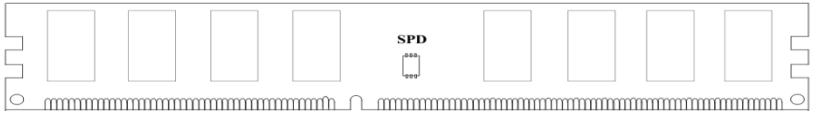
DIMM (Dual in-line memory module)



Side view



Front of DIMM



Back of DIMM

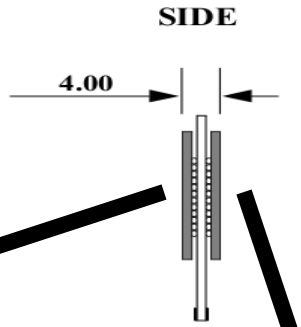


Breaking down a DIMM

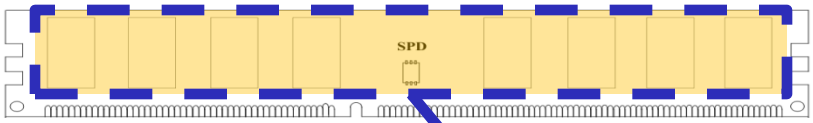
DIMM (Dual in-line memory module)



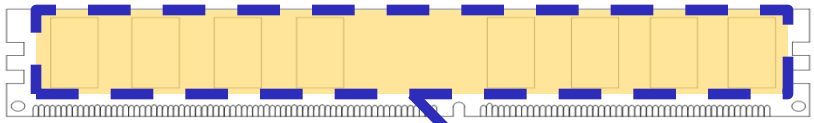
Side view



Front of DIMM



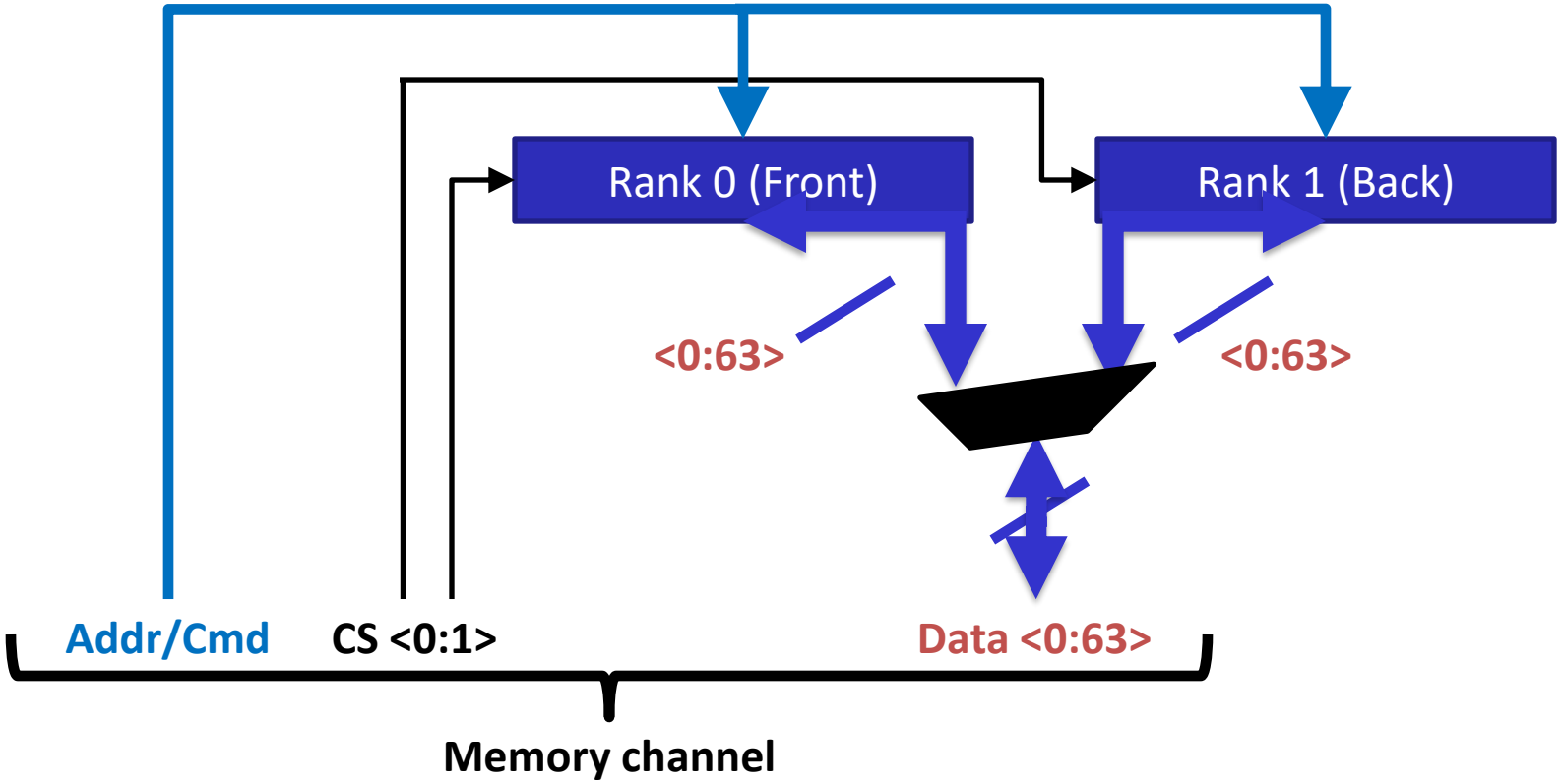
Back of DIMM



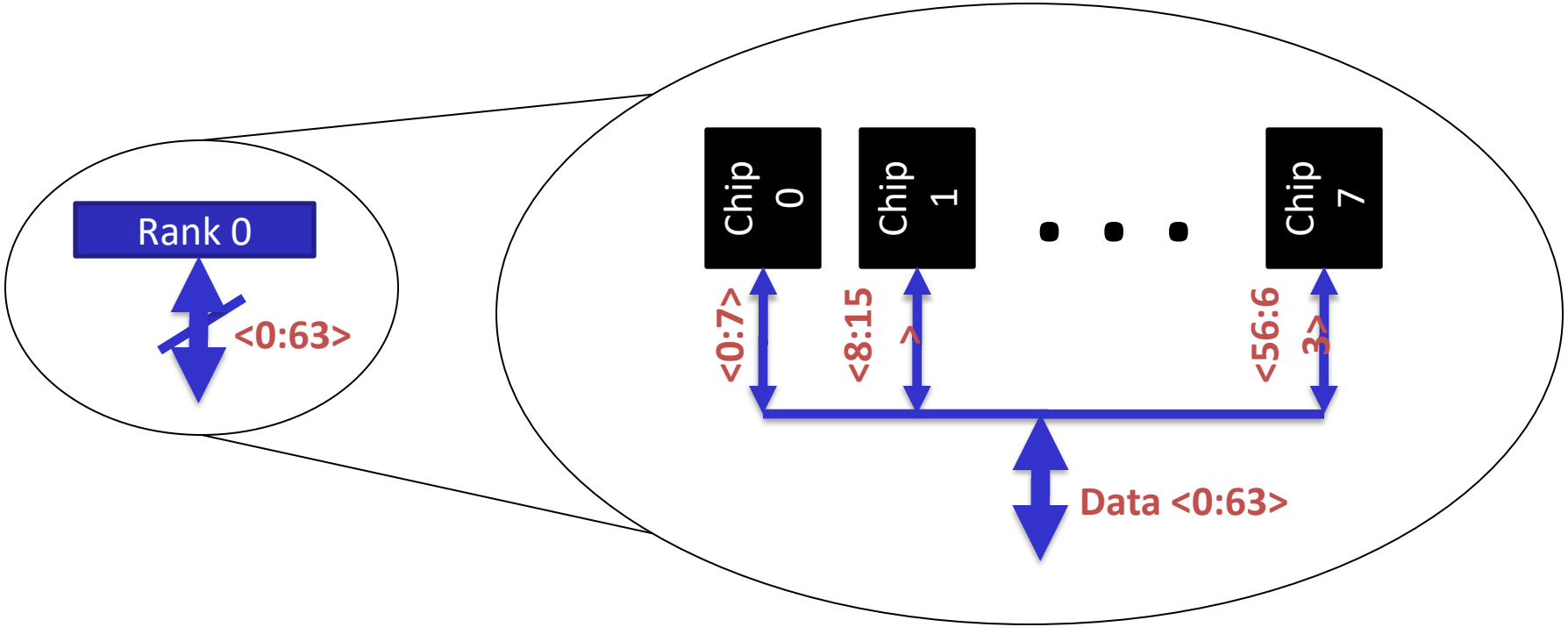
Rank 0: collection of 8 chips

Rank 1

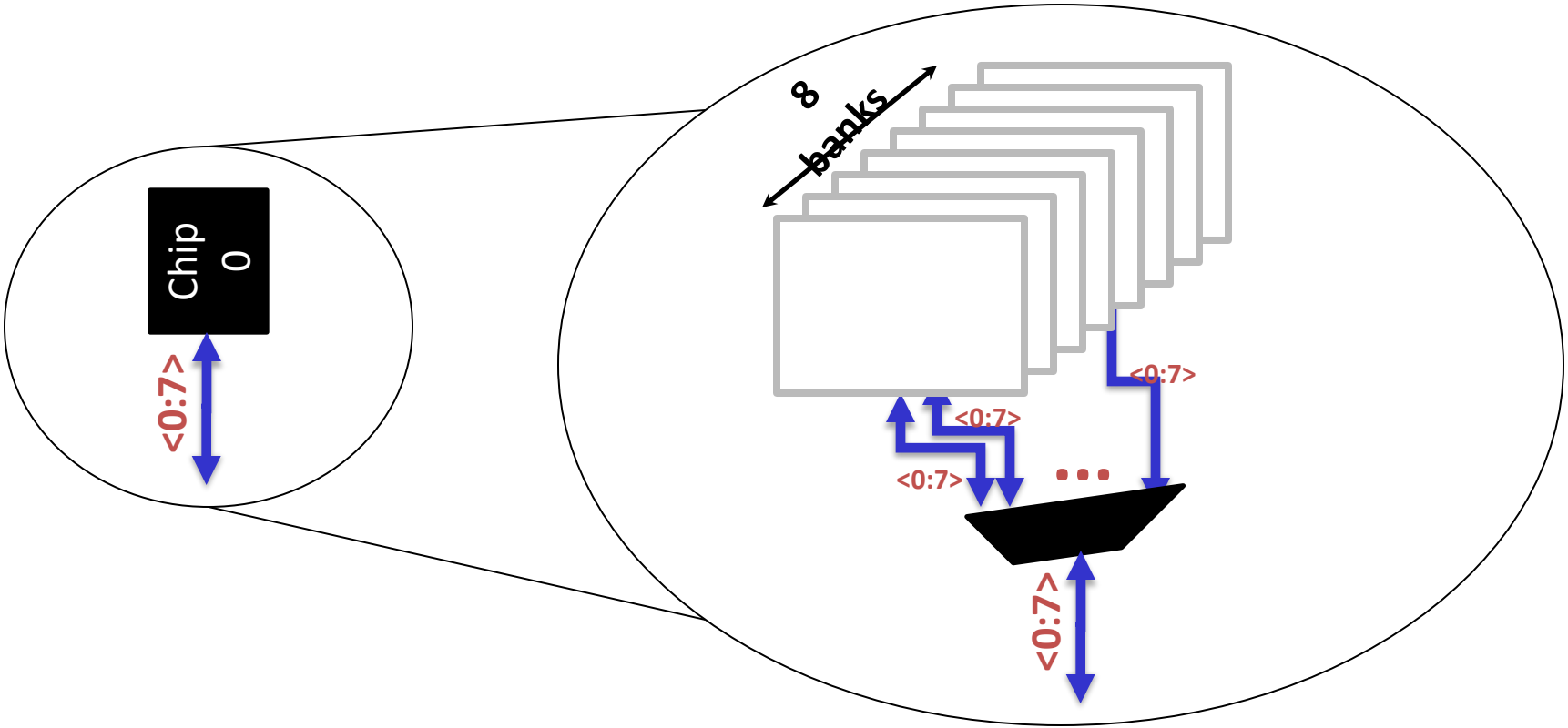
Rank


Rank 0 (Front)
Rank 1 (Back)
<0:63>
<0:63>
Addr/Cmd
CS <0:1>
Data <0:63>
Memory channel

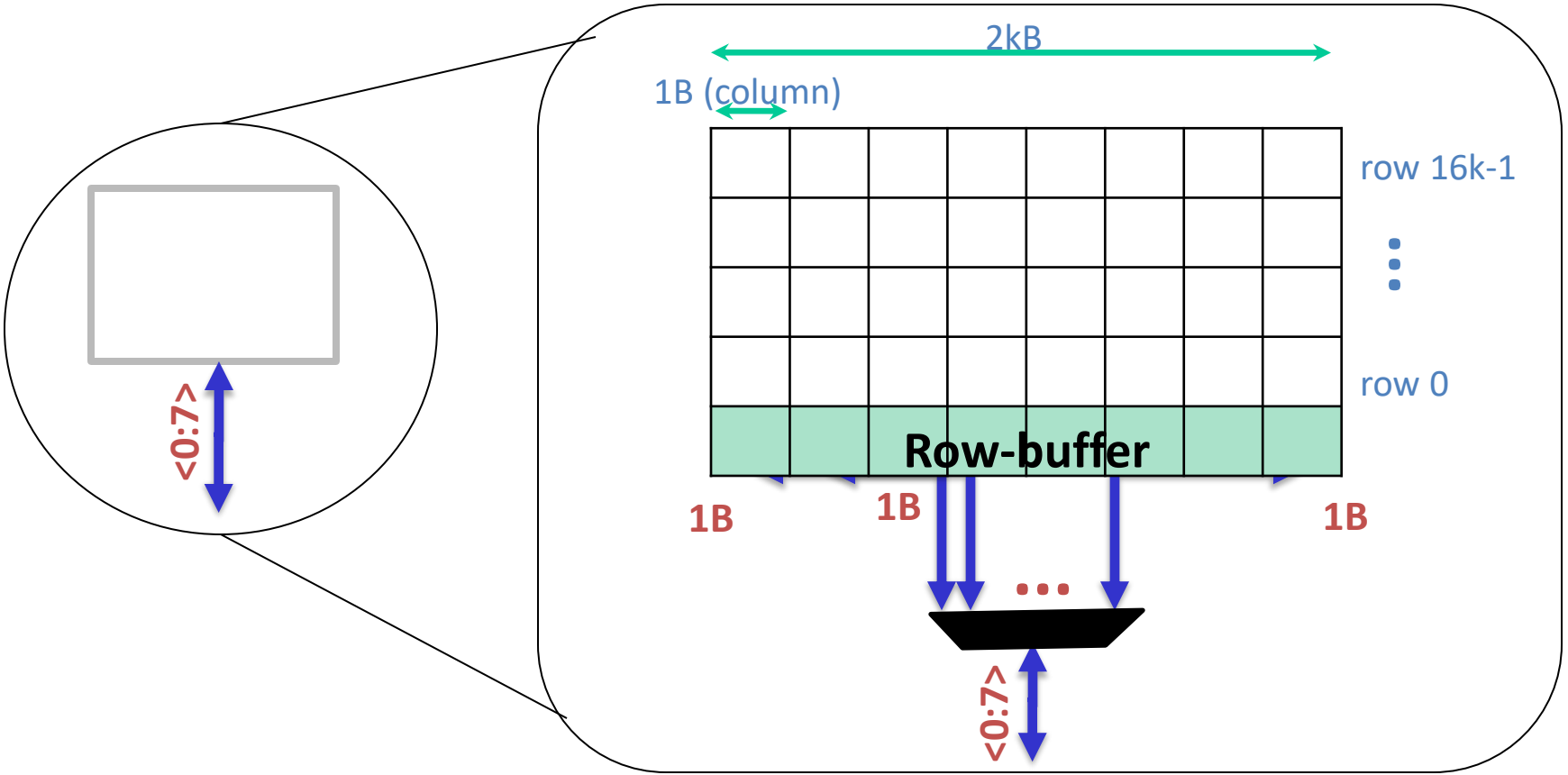
Breaking down a Rank



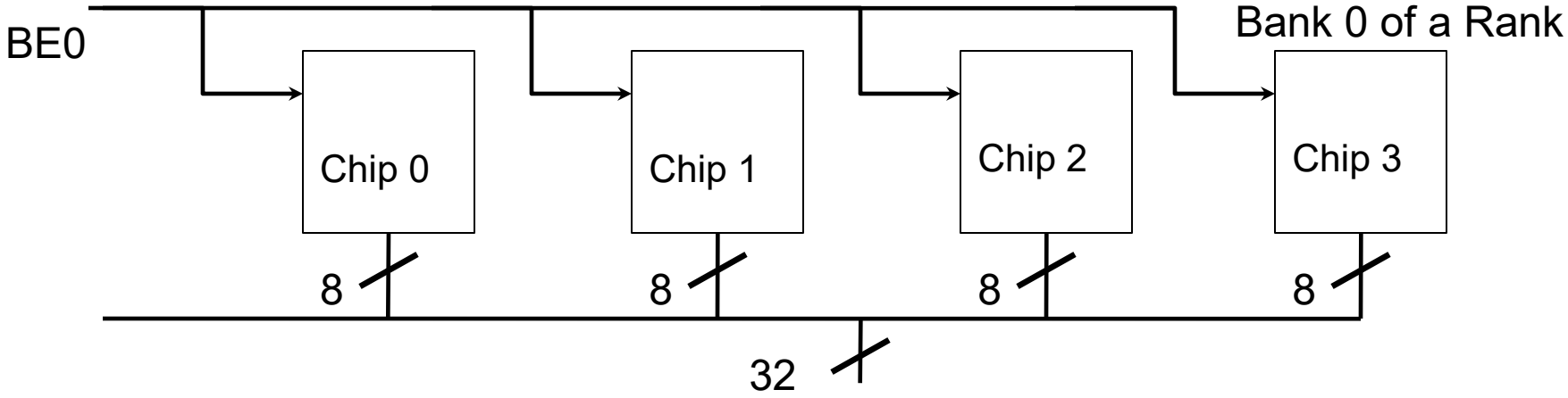
Breaking down a Chip



Breaking down a Bank



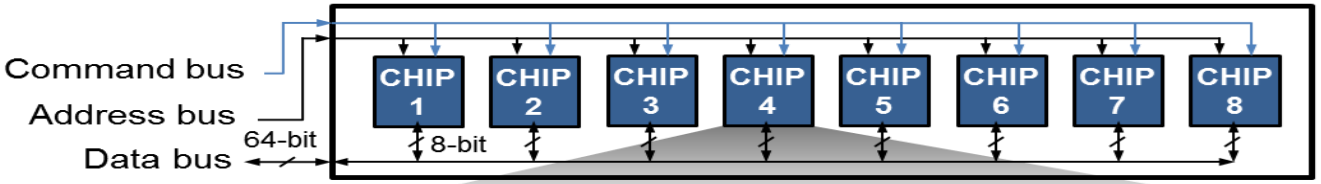
DRAM Rank



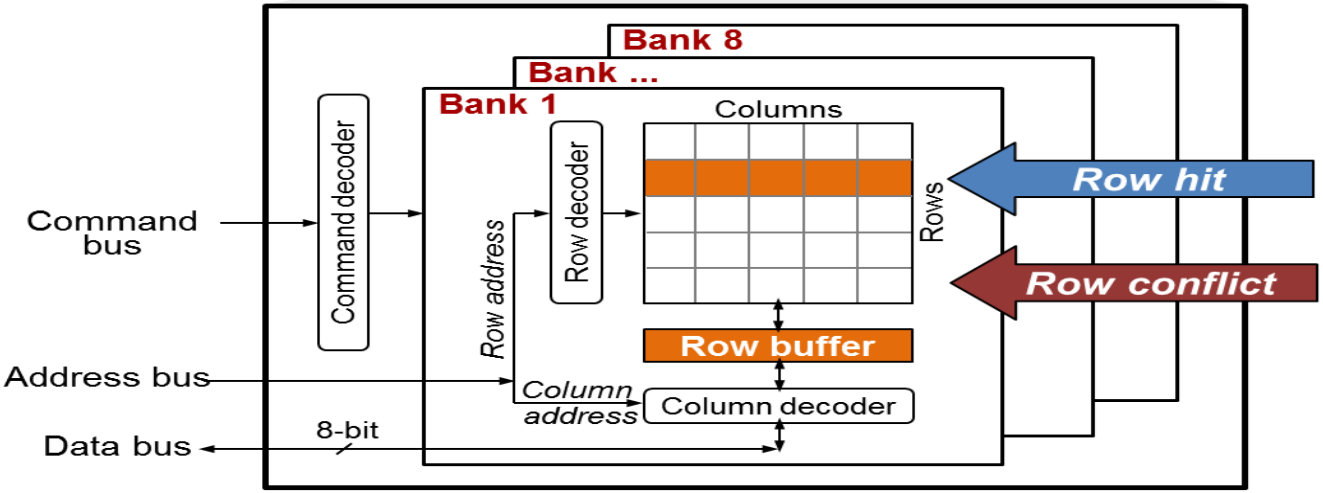
- Rank : A set of chips that respond to same command and same address at the same time but with different pieces of the requested data.
- Easy to produce 8 bit chip than 32 bit chip.
- Produce an 8 bit chip but control and operate them as a rank to get a 32 bit data in a single read.

DRAM Rank

DRAM Rank



DRAM Chip



DRAM access latency varies depending on which row is stored in the row buffer



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